Aiming for the Sun

Nigar Shaji, *59*
Project director, Aditya L1 solar mission, Indian Space Research Organisation

ACCOMPLISHMENT
Putting a spacecraft in the Halo Orbit

BY EARLY JANUARY, ISRO'S Aditya L1 mission to study the Sun will reach Lagrange point 1 (L1), around which the spacecraft, carrying seven scientific instruments, will circle in what is called the Halo Orbit. This is where the gravitational forces of the Sun and Earth nullify each other, allowing for a constant, uninterrupted view of the Sun. Solar radiation and magnetic storms can be observed before being influenced by Earth's magnetic field. Since its launch on September 2, it has travelled nearly 2.3 million km.

It has been an equally long journey for Project Director Nigar Shaji who has overseen the Aditya mission for the past eight years since its conception. "It's technologically challenging," she says. The solar mission payloads involved coordination between several scientific institutes. The halo orbit is also a tricky place to be in since the pull between the Sun and Earth needs to be balanced. "We are doing this for the first time," she says.

The first engineer from a farming family, Nigar's early motivation for a career in science came from her father Sheik Meeran. He was a mathematics graduate, even though he chose to take up farming. His stories of Marie Curie, the first woman to win a Nobel Prize, left a lasting impression on a young Nigar growing up in Shenkottai, in the foothills of the Western Ghats, on the Tamil Nadu-Kerala border. In harvest season, when the family stayed overnight on the farm, her father would tell her how the luminescence they could see was no spirit, rather it was the effect of the methane gas rising from the swampy fields.

A state rank holder in Class 10 and a district topper in Class 12, Nigar took up electronics and communication for her B.Tech. degree at a college in Tirunelveli. In 1987, a newspaper advertisement for a position at ISRO saw her joining the space agency at Sriharikota. In 1991, she moved to Bengaluru to work on spacecraft technology.

Married to a mechanical engineer who works in Dubai and with her two children now well-settled, Nigar has found both personal and professional fulfilment. Her advice to young women? "Set up a good support system so that you can devote more time to your career because you need to spend more time if you want to climb up the ladder." In Nigar's case, her mother offered that support. "She does that even now," says Nigar. ■

–Ajay Sukumaran



BANDEEP SINGH

PAPA KNOWS BEST

"If the mountain does not come to you, go to the mountain. Also, if the pistol doesn't fire, answer with the butt end"

BEST BREAKS

A trip to Italy in 2019, another to Leh. Also loves to trek, has done several trips to the Western Ghats as part of the trekking club in office